

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Technology and Engineering

M. Tech. (CE) Sem. – I (Backlog) University Examination, 2010-11

CE702: Advanced Computer Network

Date: 10.05.2011, Tuesday Time: 10:00 a.m. To 01:00 p.m. Maximum Marks: 70

Instructions:

- (1) All questions are compulsory.
- (2) Indicate clearly, the option you attempt along with its respective question Number.
- (3) Figures to right indicate marks.
- (4) Rough work is done in the last page of main supplementary; don't write anything on the question paper.

SECTION-I

Q-1 Answer the following questions. [11]

- [A] Why source-quench message is generated by router? [2]
- [B] What are the classes of IP addresses? What are the ranges of netid of each class? [2]
- [C] In which case, time-exceeded message is generated? [2]
- [D] Write down the steps of DHCP operation. [2]
- [E] What is direct and indirect delivery of packets? [3]

Q-2 Answer the following questions. [12]

- [A] What are the timers used in RIP protocol? [4]
- [B] Draw and explain message format of RIP protocol. [8]

OR

- [B] What are the differences between OSI and TCP/IP model? [8]

Q-3 Answer the following questions.(Attempt any two) [12]

- [A] By taking suitable network topology, explain how RIP makes routing table.
(Note: Explain process for only one router)
- [B] What are the security issues of IPv4 addresses and solutions of it?
- [C] By taking suitable example, explain fragmentation and reassembly in IPv4 packet with respect to MTU size=1400 bytes.

SECTION-II

Q-4 Answer the following questions. [11]

- [A] What is cumulative and selective acknowledgement? [2]
- [B] What are the types of name space? [2]
- [C] Write the comparison between IPv4 and IPv6 addresses. [2]
- [D] What is data and control connection in FTP? [2]
- [E] What are the sequences of socket system calls needed in echo client and server program? [3]

Q-5 Answer the following questions. [12]

- [A] What are the applications of Multicasting? [4]
- [B] How pseudo header is formed in TCP and UDP? What are the advantages of it? [8]

OR

- [B] What are the responsibilities of Transport Layer? How does it provide reliability of packet? [8]

Q-6 Answer the following questions. [12]

- [A] How does slow start and additive increase responds to congestion in TCP? [6]
- [B] Explain Multicast Distance Vector Routing Protocol. [6]

OR

Q-6 Answer the following questions. [12]

- [A] Explain TCP three-way handshaking. [6]
- [B] By taking suitable example, explain various presentation of IPv6 address. [6]